Master SIU Electronics Function Test Instructions

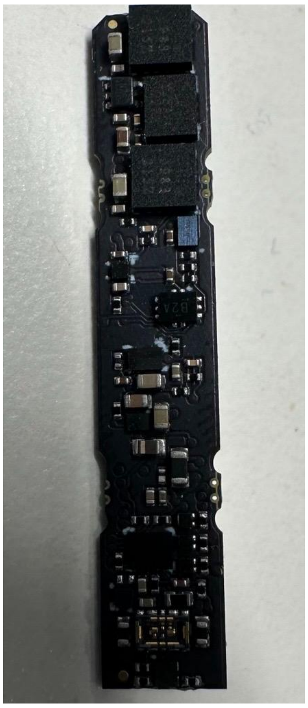
Overview:

This document contains instructions for performing a functional test on the 13506 master siu pcb. The test instructions include 4 major steps

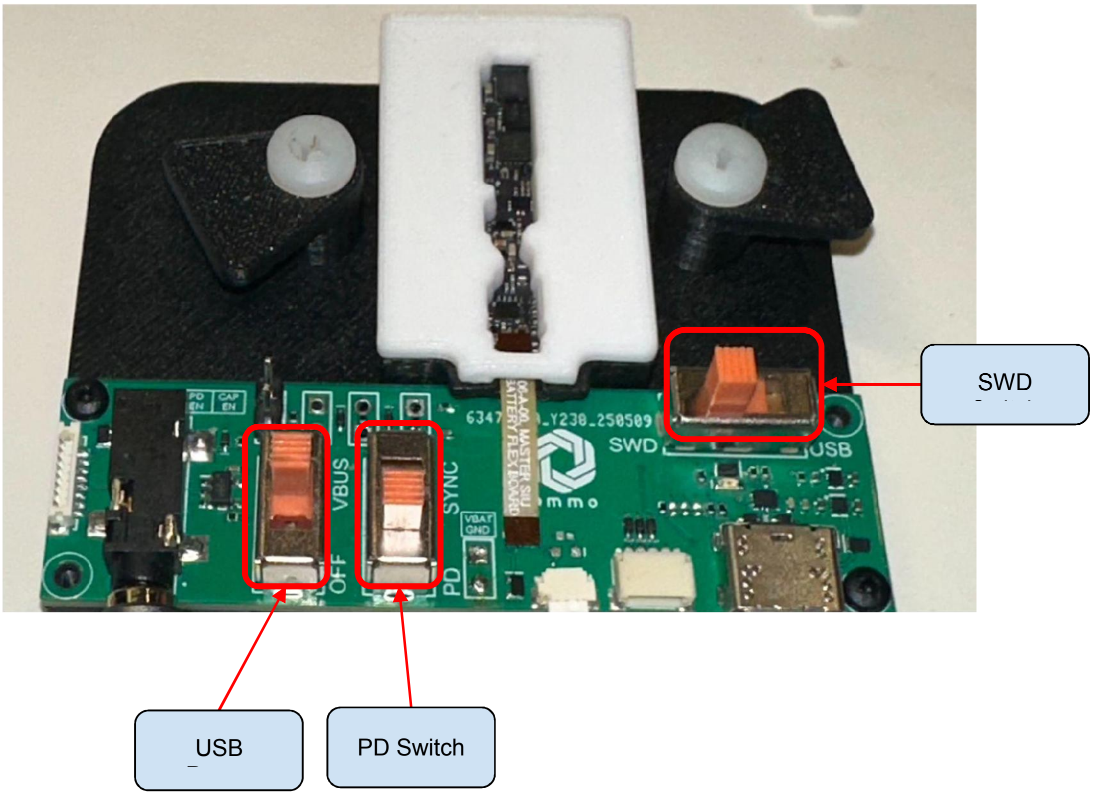
1. Program bootloader
2. Update EFT firmware
3. Perform EF Test
4. Update standard firmware

Required hardware list and description

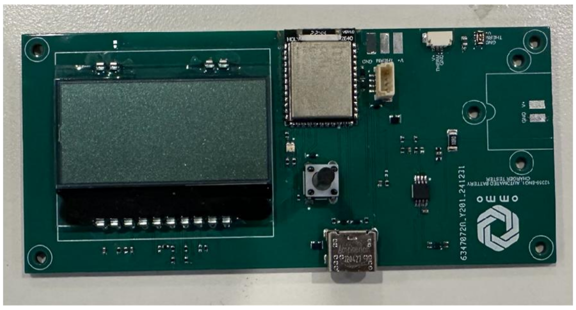
1. 13506 MSIU PCB under test



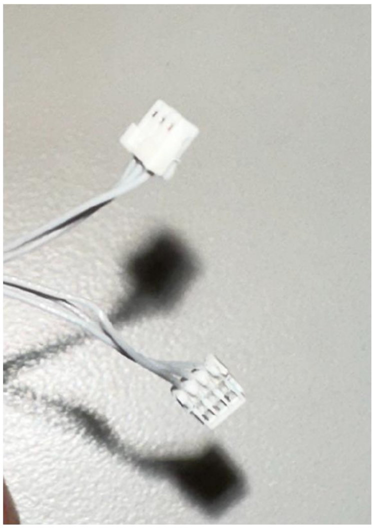
2. 13075 MSIU EFT Fixture



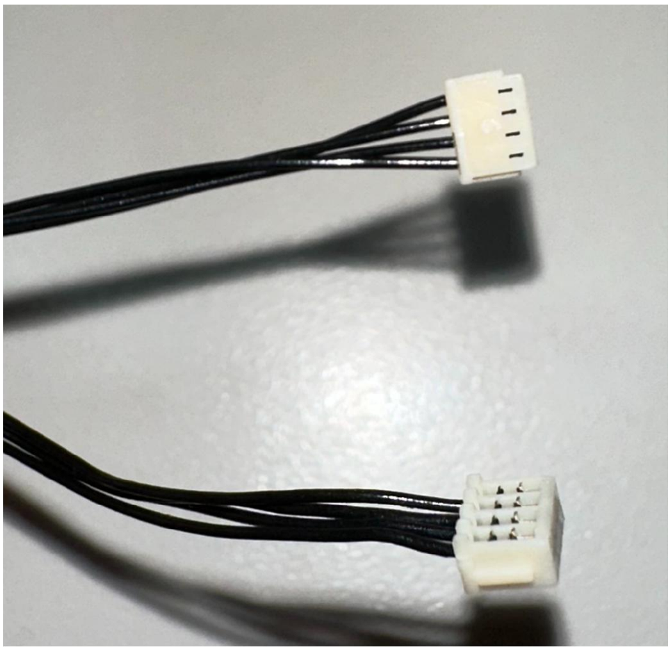
3. 12359 Charger test PCB



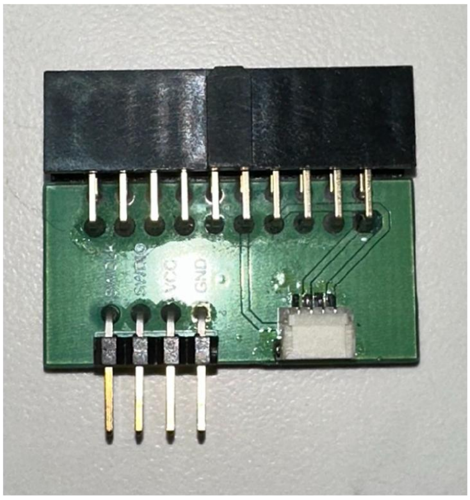
4. 3 pin jst cable



5. 4 pin JST cable



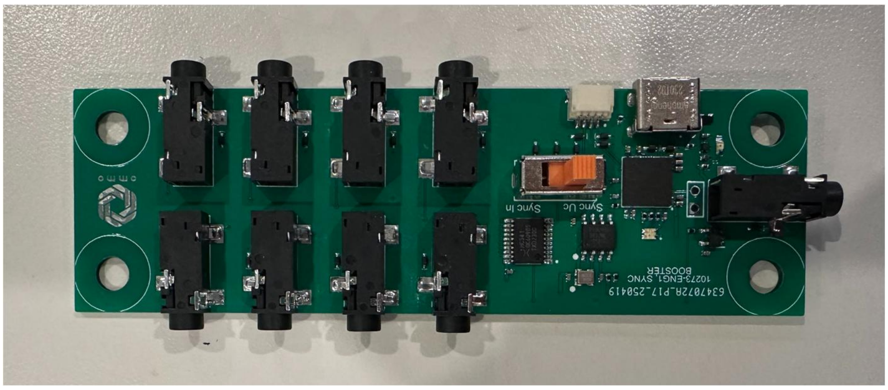
6. JLink breakout Board



7. Jlink

8. Audio Cable

9. Sync generator board



10. 3x usb-c cable
11. 13106 FPC

Software required

1. Ommo Electronics Functional Test GUI
2. Ommo Firmware Updater

Software Setup

1. Install the Ommo Electronics Functional Test GUI
2. Install nrfjprog, and nrf command line tools
    a. Download and install - https://nsscprodmedia.blob.core.windows.net/prod/software-and-other-downloads/desktop-software/nrf-command-line-tools/sw/versions-10-x-x/10-24-2/nrf-command-line-tools-10.24.2-x64.exe
    b. Once the installation is complete, verify that the following file is present at this location - `“C:/Program Files/Nordic Semiconductor/nrf-command-line-tools/bin/nrfjprog.exe”`
3. Download nrfutil.exe - (windows x64) https://www.nordicsemi.com/Products/Development-tools/nRF-Util/Download#infotabs
4. Copy nrftutil.exe to - `“C:/Program Files/Nordic Semiconductor/nrf-command-line-tools/bin/”`

5. Open the ".ini" file in - `“C:/Program Files (x86)/Ommo Technologies/Electronics Functional Test App/electronics_functional_test.ini”`
6. Ensure that the path to the above files matches the path defined in the ".ini" file -

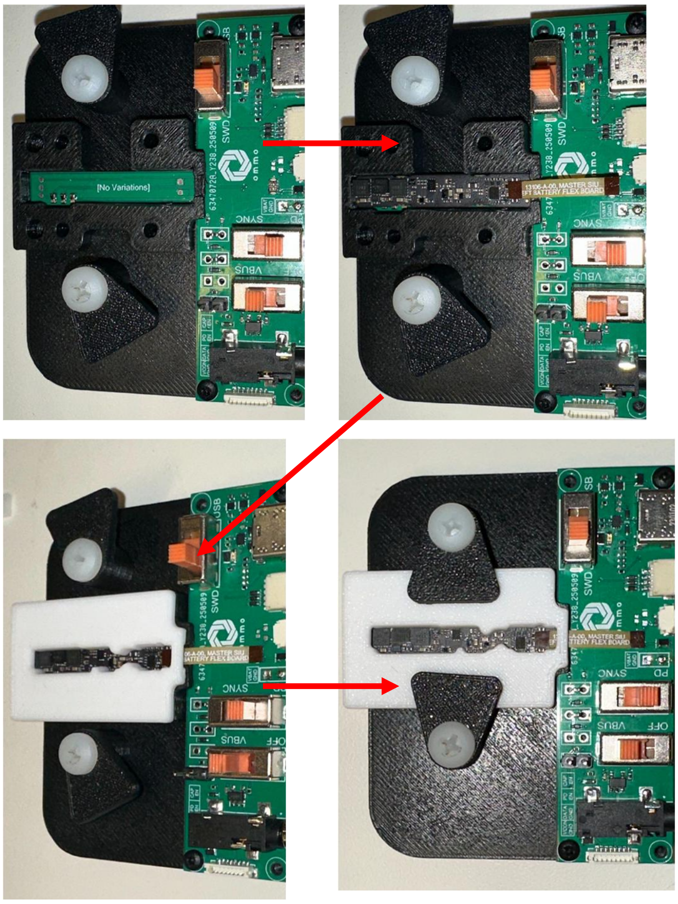
```
[DEFAULT]
log_files_directory=c:/workspace/logs/
NRFUTIL_PATH=C:/Program Files/Nordic Semiconductor/nrf-command-line-tools/bin/nrfutil.exe
NRFJPROG_ROOT=C:/Program Files/Nordic Semiconductor/nrf-command-line-tools/bin/nrfjprog
MBR_HEX=C:/workspace/ommo_nrf/nRF5_SDK_17.1.0_ddde560/components/softdevice/mbr/hex/mbr_nrf52_2.4.1_mbr.hex
```

7. Install JLink Software - https://www.segger.com/downloads/jlink/JLink_Windows_x86_64.exe
8. Make sure you have the following firmware/bootloader files -
    a. Bootloader - "13506-master_siu_2v8-bootloader-PID79-v7004_RC5.hex"
    b. EFT FW - "13506-master_siu_2v8-ef_test-3p0mm-PID79-v7004_RC5.zip"
    c. Standard FW - "13506-master_siu_2v8-standard-3p0mm-PID79-v7004_RC5.zip"
    d. Nordic MBR Hex - "mbr_nrf52_2.4.1_mbr.hex"

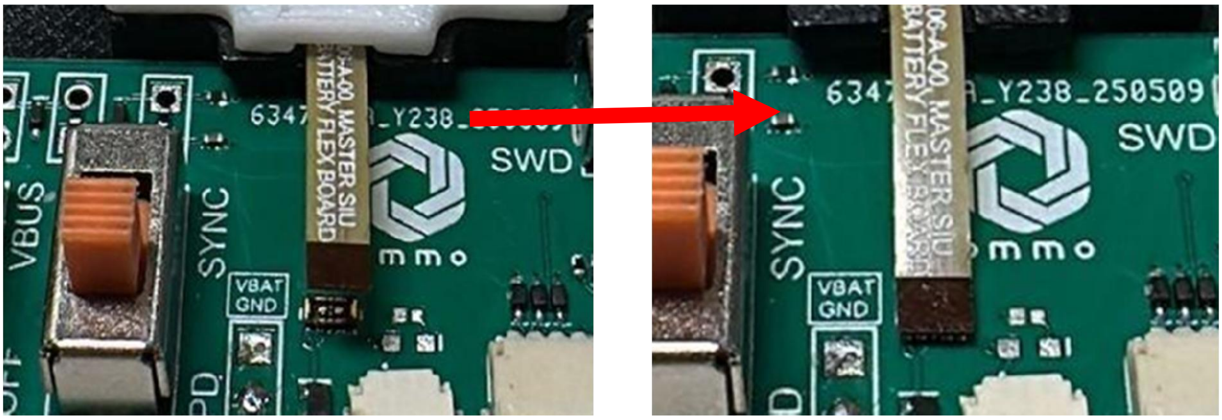
<u>Procedure</u>

Hardware connections

1. Connect the 13106 FPC to the MSIU.
2. Mount MSIU pcb on the MSIU EFT fixture (be mindful of the orientation)



3. Connect the other end of the fpc to the EFT test board.



4. Plug in 4 pin jst cable from JLink to MSIU EFT Fixture
5. Plug in the Jlink to the computer.
6. Plug in MSIU fixture to USB-C cable. **DO NOT power ON USB cable yet.**

7. Make sure the switch on the sync generator board is in the position shown below -

8. Connect one end of the audio cable to one of the sideways ports on the sync generator board.
9. Connect the other end to the MSIU EFT Fixture
10. Connect USB-C cable to the sync generator board, connect the other end to computer/usb-hub.

11. Connect the white 3 pin JST cable to the MSIU EFT fixture.
12. Connect the other end to the charger test board.
13. Connect a USB-C cable to the charger test board. **DO NOT power ON the usb cable yet.**
14. Make sure the switches on the MSIU EFT Fixture are in the following spots -

15. Power ON all USB-C connections.
16. Hardware connections are now complete.

Bootloader Programming

1. Open Ommo Electronics Functional Test App
2. Enter the “Unit Number” if you wish to track the results of the test for each board.
3. Check the “Upload Bootloader” checkbox, uncheck the rest of the checkboxes

4. Select the appropriate hex file - “13506-master_siu_2v8-bootloader-PID79-v7004_RC5.hex”
5. Click “Run”
6. The software will program the bootloader
7. You should see a message to indicate that the process has been completed.

EFT Firmware Update and Test

1. Next, switch the USB/SWD switch to USB.

2. Make sure “13506 MSIU” is selected for the SIU Type.
3. Check the “Upload Firmware” checkbox, and uncheck the “Upload Bootloader” checkbox.

4. Select the appropriate EFT firmware file - “13506-master_siu_2v8-ef_test-3p0mm-PID79-v7004_RC5.zip”
5. Click “Run”
6. The software will update the firmware
7. You should see the following messages, once the firmware is uploaded successfully.

Test procedure

1. From the dropdown, select 13506 MSIU
    a. Uncheck the “Upload EFT Firmware” checkbox.
    b. Check the “Run EF Test” checkbox

    c. Make sure the switches on the MSIU EFT Fixture are in the following positions -

    d. Click Run
    e. Follow the on-screen instruction and below descriptions to facilitate the EF Test as described below.
2. EFT Procedure
    a. Test 20-54 will run without user input.
    b. For Test 61, you will be prompted to select “Yes” or “No” for the device LED test.

        i. Visually inspect the PCB, and confirm LED colors

        ii. Click “Yes” if all three colors present
        iii. Click “No” if any one of the colors is missing
    c. Next, you will be prompted to confirm if the charger current is between 75mA and 100mA. Visually inspect the LCD screen on the charger test PCB, and confirm whether the current is between 75mA and 100 mA.

        i. Click “Yes” if the current is between 75mA and 100 mA.

        ii. Click “No” if the current value is not in the acceptable range.
    d. Next, you will be prompted to HOLD the charger board button.

        i. Press and hold the charger test PCB button. **Keep it pressed until you are prompted to release the button.**

    ii. **NOTE: the above action must be performed, even if the label on the button is "SKIP"**
    iii. Then click "SKIP" on the pop-up

e. Next, you will get a pop-up to release the charger test PCB button.

    i. Release the button

    ii. **NOTE: the above action must be performed, even if the label on the button is "SKIP"**
    iii. Click "SKIP"

f. Next, you will get a pop-up to set the USB power switch to OFF position.

    i. Switch to the expected position

    ii. **NOTE: the above action must be performed, even if the label on the button is "SKIP"**
    iii. Click "SKIP"

g. Next, you will get a pop-up to set the PD switch to SYNC position.

    i. Switch to the expected position

    ii. **NOTE: the above action must be performed, even if the label on the button is "SKIP"**
    iii. Click "SKIP"

h. The software will now perform the Clock Sync test. You will see the following message - this may take a few seconds

NOTE : For the next two tests, the MSIU will power itself OFF, and then back ON. If at any time the sequence of switches is wrong, the MSIU may be stuck in a power OFF state. If this happens, follow the steps listed in the appendix to reset the MSIU hardware, and start the test again. Also refer to the common error modes description in the appendix.

i. Next, you will be prompted to switch the PD button to PD position.

    i. Switch to the expected position

    ii. Once the above action is performed, the software will automatically move to the next step.
    iii. If you don't receive the next message in a few seconds, click "OK"

j. Next, you will be prompted to switch USB power switch to VBUS

    i. Switch to the expected position

    ii. Once the above action is performed, the software will automatically move to the next step.
    iii. If you don’t receive the next message in a few seconds, click “OK”

k. The test procedure is now complete.
l. Please check the final result displayed in the horizontal bar at the bottom.
m. If you see a green PASS, the device has passed EFT.

n. If you see a red FAIL, the device has failed EFT.

Update standard firmware on the MSIU

1. After the MSIU has passed the EF Test procedure, we need to update the firmware on it to standard sampling firmware.
2. Check the “Upload Standard Firmware” checkbox.
3. Uncheck the “Run Test” button
4. Select the appropriate standard firmware zip file - “13506-master_siu_2v8-standard-3p0mm-PID79-v7004_RC5.zip”
5. Click “Run”
6. The software will update the firmware
7. You should see the following messages.

8. Once you see the “Standard firmware upload complete” msg, this step is successfully complete.
9. Before you remove the MSIU PCB from the fixture
    a. switch the USB-SWD switch to SWD.

    b. Power OFF the USB HUB, and all USB connections.
10. The next PCB can now be tested.
11. Follow all instructions from the start, making sure to check the connections, and switch positions at each step.
12. Make sure the HUB is not powered ON until the next PCB is securely installed into the fixture.

APPENDIX

Steps to reset MSIU if it is stuck in power off/on sequence -

1. Switch the PD switch to PD
2. Click “Run Test” - you will run into an error, since the previous test is still continuing
3. Switch the USB switch to OFF
4. Click “Run Test” - you will run into an error, since the previous test is still continuing
5. You may have to click “Run Test” a third time, which will also result in an error.
6. Finally, click “Run Test” again. Once you see the different test results appear, the MSIU has been reset successfully, and you can resume testing.

Common Failure modes and their descriptions -

Following are some messages that you may encounter during the test. If you see any of these messages, it indicates that the test was not completed successfully, and needs to be performed again for that specific unit.

Case 1 -

1. This message will be displayed if the test is running, and one of the switches was not set to the correct position before clicking “OK”.
2. If you see this message, follow the instructions to reset the MSIU hardware mentioned above.
3. If you encounter this message again, and the switches have been correctly set, then consider this unit of MSIU as a EFT - “FAIL”.

Case 2 -

1. If you see this message - “EF Test Begins message not read!”, this indicates that the test did not start from the very beginning, and the MSIU is stuck in a power off/on sequence.
2. If you see this message, follow the instructions to reset the MSIU hardware mentioned above.
3. If you encounter this message again, and the switches have been correctly set, then consider this unit of MSIU as a EFT - “FAIL”.

Case 3 -

1. If you see this message,
    a. First make sure that the connections are correct as described in the setup section.

b. If you still see this message, it’s possible that the MSIU is stuck in a power on/off sequence.

2. Follow the instructions to reset the MSIU hardware mentioned above.

Case 4 -

1. If at any point, the software is waiting on this screen for more than a few seconds, please wait for some time to allow the software to reset with the following error message -
2. The next time you click “Run Test” the software will resume functionality.